PP2

Intelligent Betel Leaf Management System

for Disease detection, Leaf Quality Grading, Yield Prediction and Pest detection


Disease Detection


Leaf Quality Grading


Yield Prediction


Pest Detection

Research Progress Presentation 2

R62-IT-062



RESEARCH BACKGROUND

Problem -

Betel leaf farmers mainly rely on manual inspection and personal experience to assess leaf quality, identify diseases and pests, and estimate yield. This makes the process time-consuming, subjective, and difficult to perform consistently.

Effect -

These challenges can lead to inaccurate quality grading, delayed disease and pest identification, crop damage, reduced leaf quality and yield, and difficulty in making informed harvesting and farming decisions.

Solution -

Our project develops an AI-based mobile application that integrates quality grading, disease detection, pest detection, and yield prediction, allowing farmers to analyse leaf images and enter field measurements to receive faster and data-driven farming insights.



Intercropping & Betel Research Station (IBRS), Narammala

Components

- Quality Grading
- Disease Detection
- Yield Prediction
- Pest Detection



FULL SYSTEM WORKFLOW

AI-Powered End-to-End Betel Leaf Management & Advisory Platform

STAKEHOLDERS

Farmers
Data Entry, Insight Access, Decision Making

Exporters & Suppliers
Quality Verification, Export Compliance, Product Tracking

Agricultural Experts
Monitoring, Advisories, Recommendations

Government & Institutions
Policy, Support, Analytics, Oversight

1 Data Collection

Image Data
- Leaf Images
- Pest Images

Environmental Data
- Temperature
- Humidity
- Soil Moisture
- pH, NPK

Manual Entries
- Farmer Input
- Geotags
- Timestamps

External Data
- Weather API
- Rainfall Data
- Historical Records

2 AI Processing & Models

1 Quality Grading & Export Verification
EfficientNet-B0
+ OOD Detection

2 Disease Detection & Management
EfficientNet-B0
+ OOD Detection

3 Pest Detection & Management
YOLOv8m
+ OOD Detection

4 Yield Prediction & Micro-Climate Optimization
LSTM + Random Forest + XGBoost

3 Decision & Advisory

Quality Decision
Quality Grade
Export Acceptability

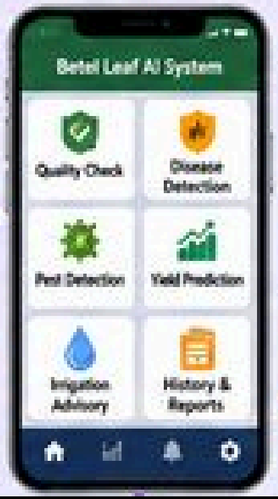
Disease Advisory
Disease Type
Treatment Suggestion

Pest Control Advisory
Pest Type
Integrated Management

Irrigation & Yield Advisory
Daily Watering Plan
Yield Forecast (30–60 Days)

4 Delivery & Access

Mobile Application (Flutter)



5 Outcomes & Impact

- Higher Quality Export Standards
- Early Disease & Pest Detection
- Optimized Irrigation & Resource Use
- Increased Yield & Profitability
- Reduced Chemical Usage
- Data-Driven Farming Decisions

AI-Based BETEL LEAF

QUALITY GRADING AND EXPORT VERIFICATION

- AI-Powered Leaf Analysis
- Quality Grading
- Export Verification
- Reliable & Accurate



IT NUMBER	IT22290892
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DEGREE	B.Sc. (Hons) in Information Technology

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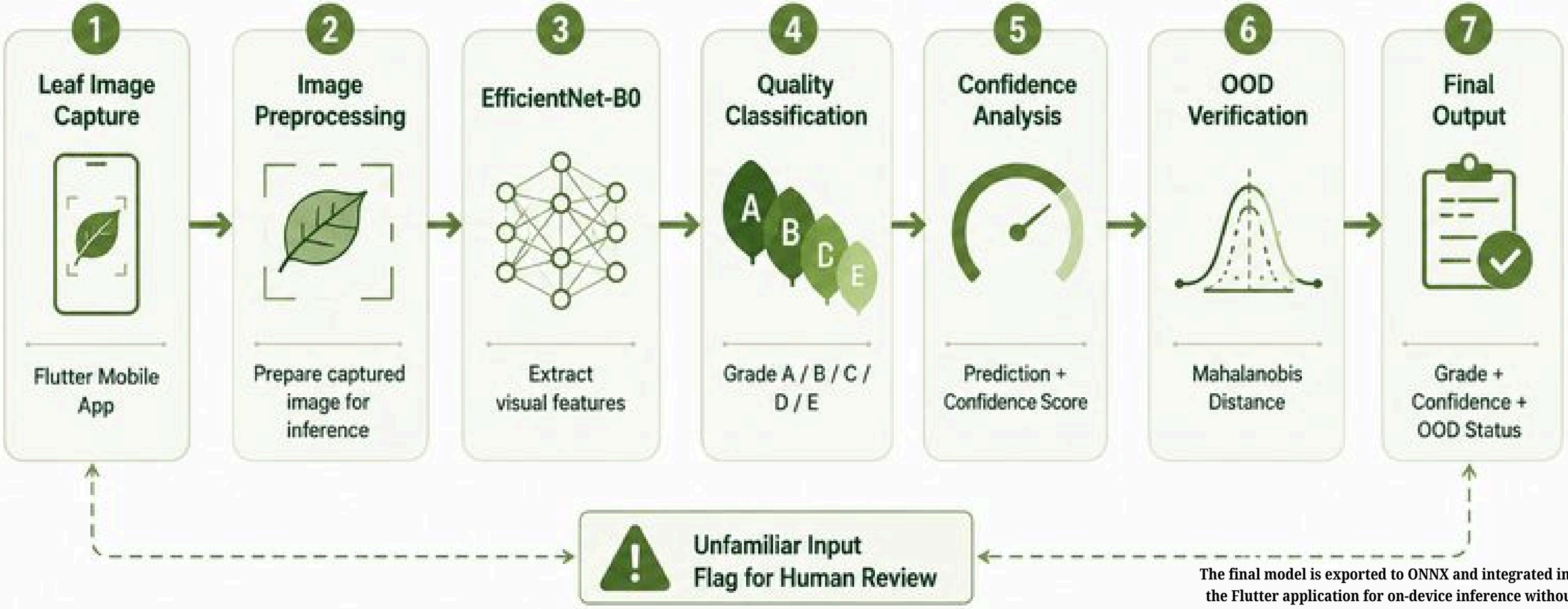
RECOMMENDATION AND COLLABORATION



- Addresses the problem of **manual and subjective betel leaf quality grading** before export.
- Classifies betel leaves into **five quality grades (A–E)** based on visual quality.
- Uses a fine-tuned **EfficientNet-B0** deep learning model for image classification.
- Uses **Mahalanobis-distance Out-of-Distribution (OOD)** detection to identify unfamiliar inputs.
- Provides **grade prediction, confidence score and OOD status** through a Flutter mobile application.
- Designed for practical use by **packing-house operators** during pre-export quality verification.

Novelty of the Component,
Combines AI-based five-grade betel leaf classification with OOD detection and mobile deployment for reliable export quality verification.

QUALITY GRADING & EXPORT VERIFICATION WORKFLOW



The final model is exported to ONNX and integrated into the Flutter application for on-device inference without requiring a server connection.

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RECOMMENDATION AND COLLABORATION



PP1 Feedback

- Model training only.
- No GUI implementation.
- The complete system was expected to reach 90% completion by the PP2 milestone.

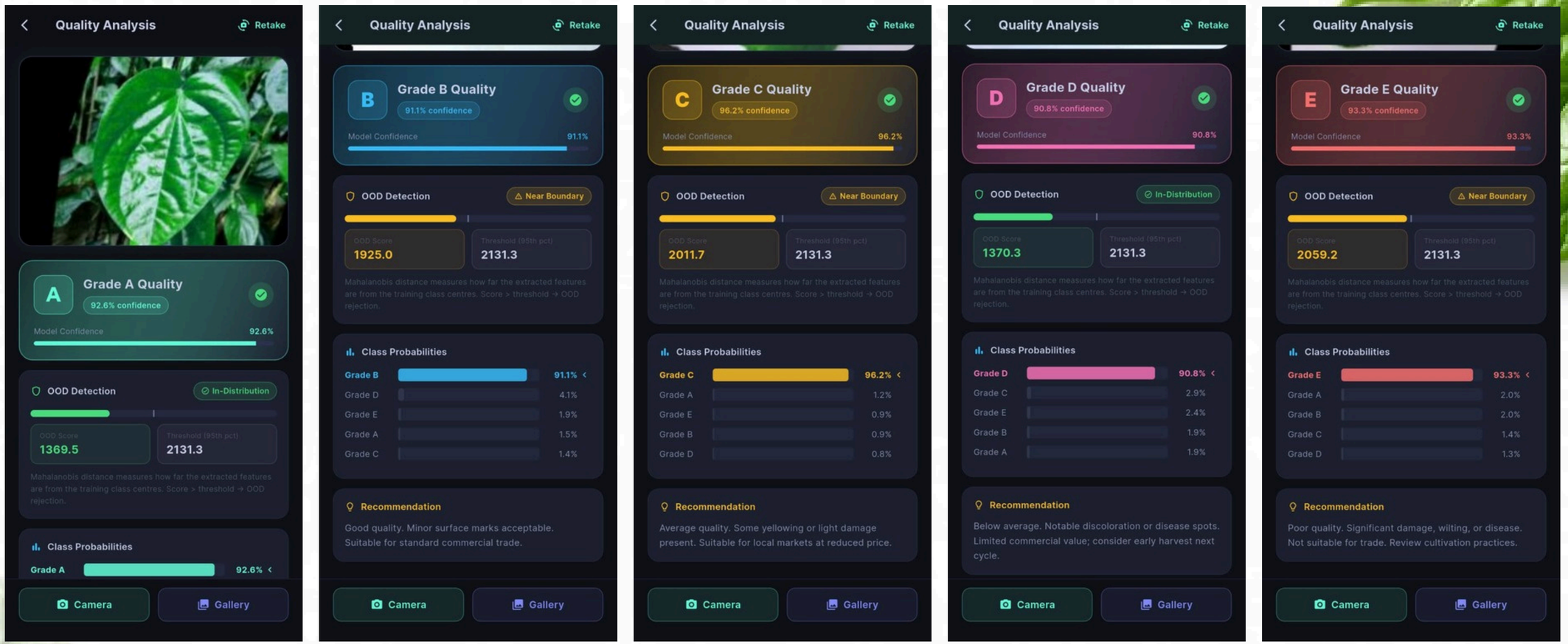
PP2 Improvement

- Completed the Flutter mobile application.
- Integrated the trained EfficientNet-B0 model into the application.
- Implemented Grade A-E quality classification.
- Added confidence score and OOD verification.
- Completed the end-to-end workflow from image capture to quality verification.
- System now supports on-device inference for practical use.

PP1: AI Model Only → PP2: Complete End to End Mobile System

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RESULTS AND DISCUSSION



The completed system demonstrates accurate five-grade betel leaf classification with OOD-based verification and practical on-device deployment through the Flutter mobile application.

AI SYSTEM for

BETEL LEAF

YIELD PREDICTION AND MICRO-CLIMATE OPTIMIZATION

AI-Powered Yield Prediction

Micro-Climate Optimization

Smart Monitoring

Data-Driven Decision Making



TEAM ID	:	IT22207890
MEMBER NAMES	:	Perera A L S R
DEGREE	:	Bsc. (Hons) in Information Technology
PRESENTATION DATE	:	1/9/2026

RECOMMENDATION AND COLLABORATION



- Addresses the lack of **data-driven yield forecasting and micro-climate management** in betel cultivation.
- Uses **farmer-reported environmental data** instead of expensive fixed IoT sensor networks.
- Collects **temperature, humidity, soil moisture, pH, and NPK** readings through a Flutter mobile application.
- Uses **LSTM** for betel leaf yield forecasting.
- Uses **Random Forest and XGBoost** for environmental feature analysis and cross-validation.
- Provides **irrigation recommendations** based on soil moisture, temperature, and rainfall forecasts.
- Designed as a **low-cost and accessible solution** for smallholder betel farmers.

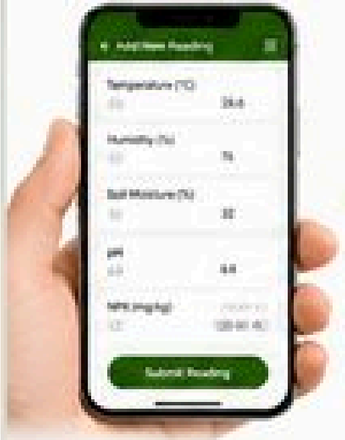
Novelty of the Component,
Combines farmer-reported environmental data with AI-based yield prediction and micro-climate optimization, eliminating the need for costly fixed IoT infrastructure.

YIELD PREDICTION & MICRO-CLIMATE OPTIMIZATION WORKFLOW

Farmer-Reported Data → AI Models → Smart Decisions → Better Betel Yield

1 Environmental Data Entry

Flutter Mobile App



Temperature • Humidity
Soil Moisture • pH • NPK

2 Data Validation & Storage

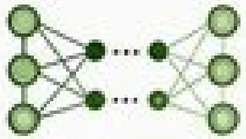
Validate + Timestamp + Geotag

3 Backend Processing

FastAPI + PostgreSQL



4 LSTM Yield Prediction



14-Day Input Sequence

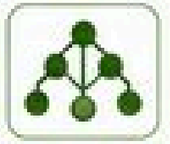
D-13 D-12 ··· D-1 D0



30–60 Day Yield Forecast

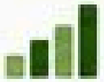
5 Feature Analysis

Random Forest + XGBoost



XGBoost

Identify Key Environmental Factors



6 Micro-Climate Optimization

Irrigation Advisory



Soil Moisture

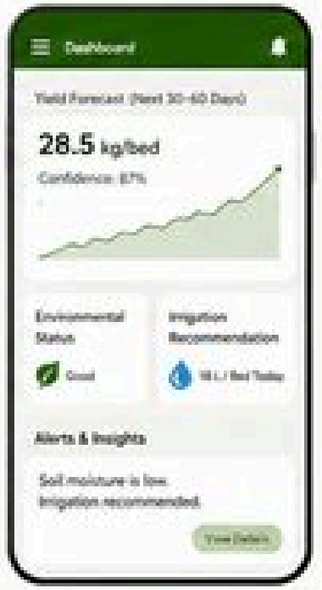
Temperature

Rainfall Forecast

Generate Daily Watering Recommendation

7 Farmer Dashboard

Forecast + Insights + Recommendations





Smart Data, Stronger Roots, Higher Yields.
AI-Powered Insights for Sustainable Betel Cultivation.

The final model is exported to ONNX and integrated into the Flutter application for on-device inference without requiring a server connection.

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RECOMMENDATION AND COLLABORATION



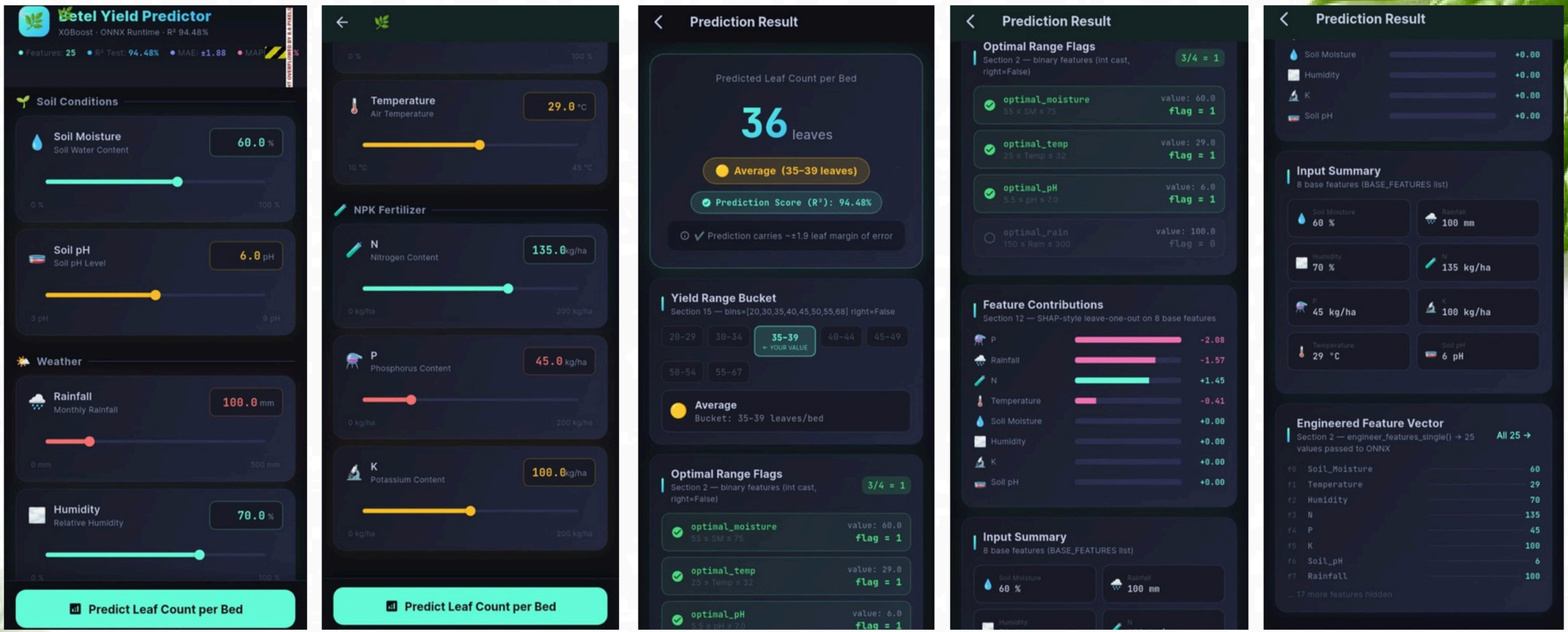
PP1 Feedback

- Round up the yield prediction result for clearer presentation.
- Add a confidence score to the yield prediction.
- System should reach 90% completion by PP2.

PP2 Improvement

- Updated yield predictions to provide clear rounded results.
- Added confidence information to improve prediction reliability.
- Developed the Flutter-based farmer interface.
- Implemented environmental data collection and validation.
- Integrated the LSTM yield prediction pipeline.
- Added Random Forest & XGBoost for environmental feature analysis.
- Developed the micro-climate and irrigation advisory module.
- Integrated the major components into the end-to-end system.

RESULTS AND DISCUSSION



The completed system demonstrates accurate five-grade betel leaf classification with OOD-based verification and practical on-device deployment through the Flutter mobile application.

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AI-POWERED

BETEL LEAF

DISEASE DETECTION

and Management Advisory System



AI-Powered Disease Detection

Management Advisory

Smart Monitoring

Healthier Leaves Better Yield



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DEGREE	:	Bsc. (Hons) IT
PRESENTATION DATE	:	1/9/2026

RECOMMENDATION AND COLLABORATION



- Detects betel leaf diseases using **AI-based image classification.**
- Uses **EfficientNet-B0** with transfer learning.
- Classifies leaves into 4 conditions:

1. Bacterial Leaf Disease
2. Fungal Brown Spot Disease
3. Dried Leaf
4. Healthy Leaf

- Uses **OOD Detection** to reject unreliable/non-leaf inputs.
- Designed for **offline mobile disease screening**.

Novelty of the Component,
Combines AI-based betel leaf disease detection with OOD verification and offline mobile deployment for reliable field-level disease screening.

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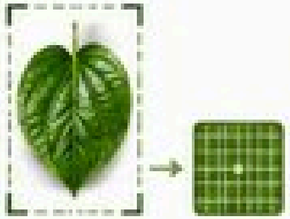
DISEASE DETECTION WORKFLOW

1
Leaf Image Capture



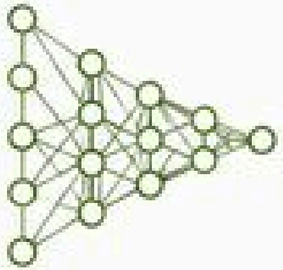
Mobile Application

2
Image Preprocessing



Resize to 224 × 224 & Normalize

3
EfficientNet-B0



Extract Leaf Features (1280-D Feature)

4
Disease Classification

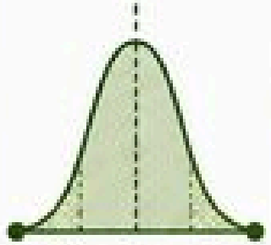
- Bacterial Leaf Disease
- Fungal Brown Spot Disease
- Dried Leaf
- Healthy Leaf

5
Confidence Analysis



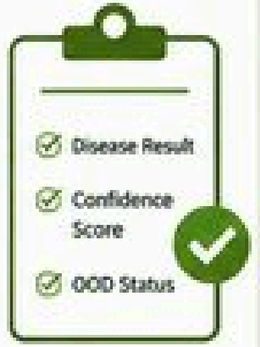
Prediction + Confidence Score

6
OOD Verification



Mahalanobis Distance

7
Final Output



Reliable AI-Powered Diagnosis for Better Betel Leaf Health

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RECOMMENDATION AND COLLABORATION

PP1 Feedback

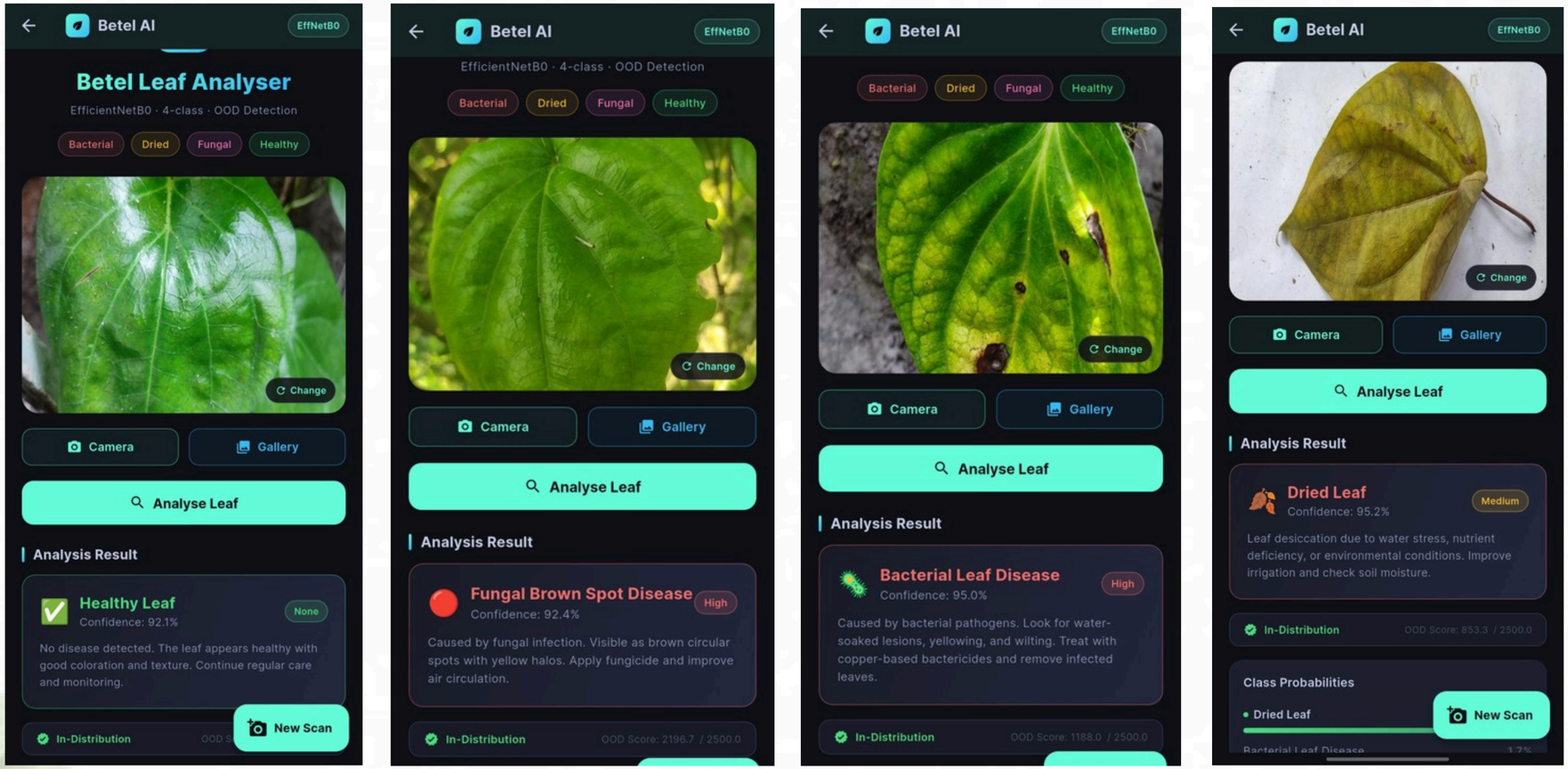
- Component was functionally working.
- Candidate needed to improve the ability to explain the code and implementation.
- System was expected to reach 90% completion by PP2.

PP2 Improvement

- Improved understanding and explanation of the complete AI pipeline.
- Completed two-phase EfficientNet-B0 training and fine-tuning.
- Implemented 4-class disease classification.
- Added Mahalanobis OOD detection.
- Exported classifier and feature extractor to ONNX.
- Prepared the model for offline mobile deployment.

Working Model → PP2: Complete, Explainable & Deployment-Ready AI Pipeline

RESULTS AND DISCUSSION



The EfficientNet-B0 model achieved high and balanced performance across all four betel leaf conditions, with OOD detection providing an additional reliability layer for invalid inputs.

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AI POWERED for

BETEL VINE

PEST DETECTION AND INTEGRATED PEST MANAGEMENT SYSTEM

AI-Powered Pest Detection

Accurate Identification

Integrated Pest Management

Smart Recommendations



TEAM ID	:	IT22165602
MEMBER NAMES	:	Fernando M N S
DEGREE	:	Bsc. (Hons) IT
PRESENTATION DATE	:	1/9/2026

RECOMMENDATION AND COLLABORATION



- Addresses the difficulty of **manual pest identification** in betel cultivation.
- Detects and localizes pests directly from betel vine images.
- Uses **YOLOv8m** for real-time object detection.
- Identifies **4 pest classes**: Green Leafhopper, Aphid, Beetle, and Grasshopper.
- Uses **Mahalanobis OOD Detection** to flag unfamiliar inputs.
- Supports an offline-capable deployment pipeline through **ONNX**.

Novelty of the Component,
Combines multi-class betel pest localization with OOD detection for reliable AI-assisted pest identification.

PEST DETECTION WORKFLOW

1 Image Capture

Capture

Betel Vine / Leaf Image



2 Image Preprocessing

Resize to 640 × 640



3 YOLOv8m Detection

Extract Features & Detect Objects

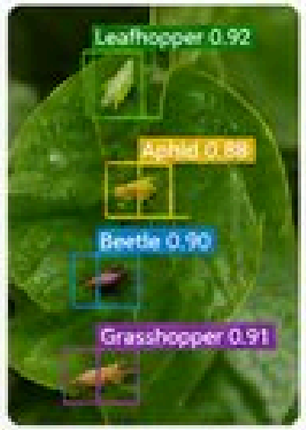


4 Pest Classification

- Leafhopper
- Aphid
- Beetle
- Grasshopper

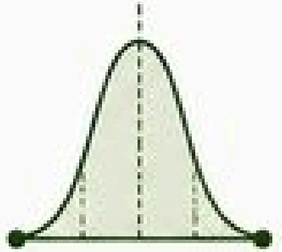
5 Pest Localization

Bounding Box + Confidence Score



6 OOD Verification

Mahalanobis Distance

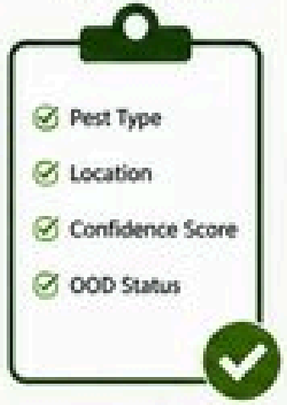


In-Distribution

Out-of-Distribution

7 Final Output

Pest Type + Location + Confidence + OOD Status



AI-Powered Pest Detection for Healthy Betel Vine

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RECOMMENDATION AND COLLABORATION

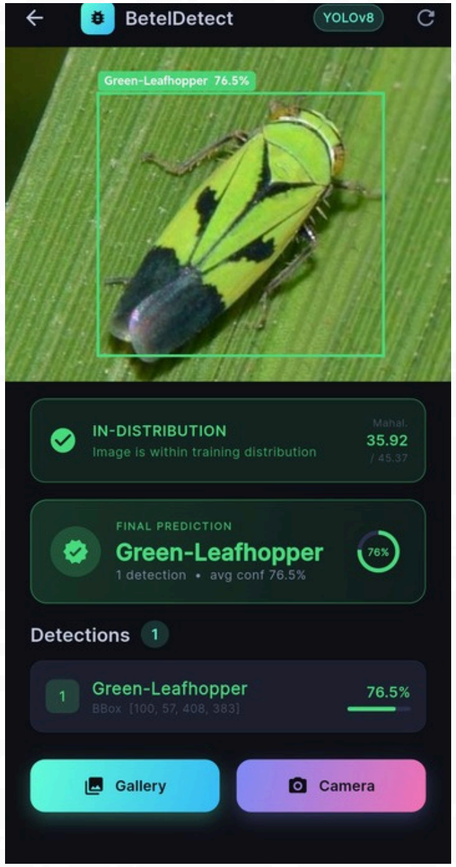
PP1 Feedback

- Component was functionally working.
- Candidate was unable to clearly explain the code to the panel.
- System was expected to reach 90% completion by PP2.

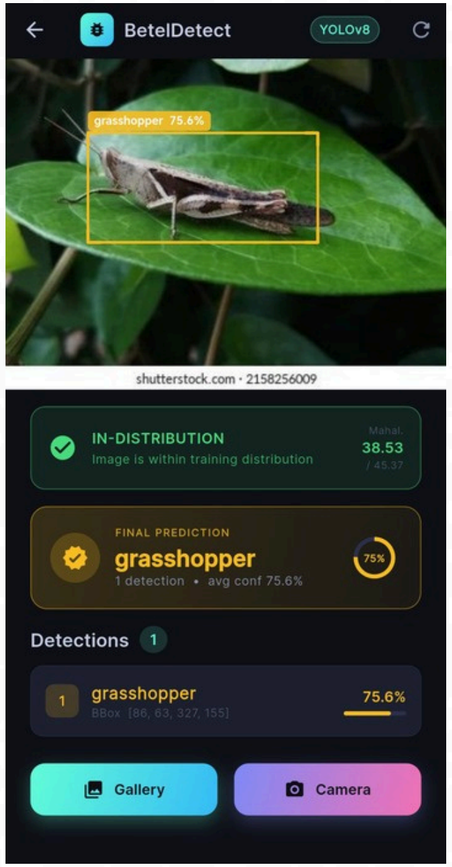
PP2 Improvement

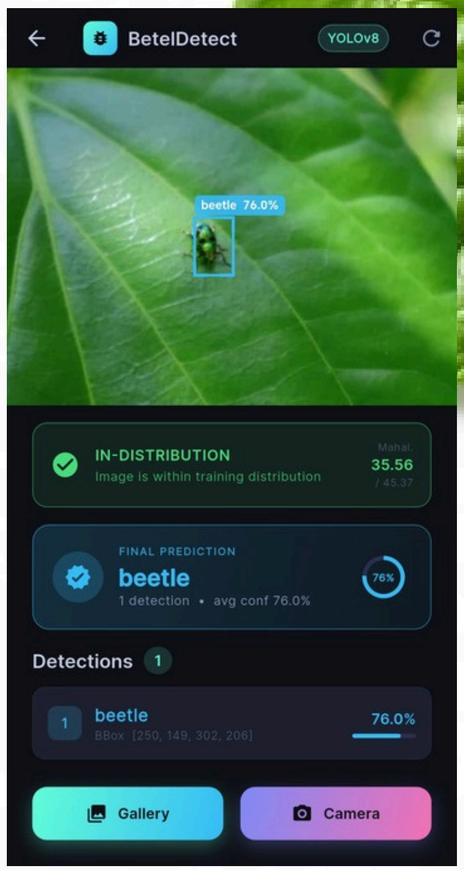
- Improved understanding of the complete model architecture and code flow.
- Finalized YOLOv8m pest detection model.
- Implemented detection for 4 pest classes.
- Added pest localization using bounding boxes.
- Exported the complete detection pipeline to ONNX.
- Prepared the component for integration with the broader Integrated Pest Management system.

RESULTS AND DISCUSSION









The YOLOv8m model achieved high pest-detection performance across four betel pest classes, while OOD verification provides an additional reliability layer for unfamiliar field inputs.

BUSINESS POTENTIAL, COMMERCIALIZATION & ETHICAL CONSIDERATIONS



COMMERCIALISATION & MARKET POTENTIAL

- Target Audience: Betel farmers, packing houses, exporters, and agricultural organisations.
- Value Proposition: One AI-powered platform for quality grading, disease detection, pest management, yield prediction, and microclimate optimisation.
- Business Model: Subscription or licensing model for farmers, farming organisations, packing houses, and exporters.

ETHICAL & SOCIAL CONSIDERATIONS

- Data Privacy: Secure handling of farmer, farm, and crop data.
- AI Reliability: AI provides decision support; important decisions should allow human verification.
- Responsible Agriculture: Promote safe and sustainable farming and pest-management practices.


THANK YOU....